WORKED SOLUTION

STEP II 2020, Question 2

Pure • Separable Differential Equations

STEP II 2020, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The differential equation separates, and after integrating, the natural thing to look at is the product xy, which turns out to be a power of 2. The identity $(x+y)^2-(x-y)^2=4xy$ then rewrites everything in terms of the sum $s=x+y$ and the difference $d=x-y$. In those variables each curve becomes "d^2 equals a function of s", and the sketches come from asking where that function is non-negative.

Solution

Part (i)

Factorise the right-hand side:

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{y(kx-1)}{x(1-ky)} \quad\Longrightarrow\quad \frac{1-ky}{y}\,\mathrm{d}y=\frac{kx-1}{x}\,\mathrm{d}x.$$

Integrating, $\ln|y|-ky=kx-\ln|x|+c$, that is

$$\ln|xy|=k(x+y)+c.$$

On both curves x and y have the same sign, so $|xy|=xy$. Since $k=\ln 2$, exponentiating gives $xy=A\,2^{x+y}$ for a positive constant A.

Curve C_1. At $(1,1)$ we get $1=4A$, so $A=\frac{1}{4}$ and $4xy=2^{x+y}$. Now $4xy=(x+y)^2-(x-y)^2$, so

$$(x-y)^2=(x+y)^2-2^{x+y},$$

with $x,y>0$. This is the required result.

Curve C_2. At $(-1,-1)$ we get $1=A\cdot 2^{-2}$, so $A=4$ and $4xy=16\cdot 2^{x+y}=2^{x+y+4}$. Hence

$$(x-y)^2=(x+y)^2-2^{x+y+4},$$

with $x,y<0$.

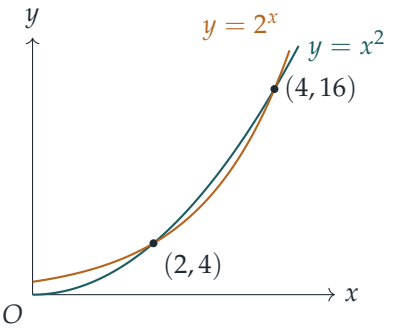
Symmetry. Swapping x and y leaves $x+y$ unchanged and leaves $(x-y)^2 = (y-x)^2$ unchanged, so each equation is unchanged. The sign conditions ($x, y > 0$, or $x, y < 0$) are unchanged too. So whenever (a, b) lies on a curve, so does (b, a), its reflection in $y = x$. Hence $y = x$ is a line of symmetry of each curve.

Part (ii)

For $x \geq 0$, the curves $y = x^2$ and $y = 2^x$ meet at $(2, 4)$ and $(4, 16)$. These are the only meeting points. To see this, compare logarithms: let $g(x) = 2\ln x - x\ln 2$ for $x > 0$. Then $g'(x) = \frac{2}{x} - \ln 2$, so g increases up to $x = 2/\ln 2$ and decreases after it. So g has at most two zeros, and $g(2) = g(4) = 0$. Hence

$$x^2 \geq 2^x \iff 2 \leq x \leq 4 \qquad (x > 0),$$

and 2^x lies above x^2 on $[0, 2)$ and on $(4, \infty)$.



Now take a point of C_1 and put $s = x + y > 0$. The left-hand side of the equation of C_1 is a square, so

$$s^2 - 2^s = (x-y)^2 \geq 0, \qquad \text{that is} \qquad s^2 \geq 2^s.$$

By the sketch, this forces $2 \leq s \leq 4$. So C_1 lies between the lines $x + y = 2$ and $x + y = 4$.

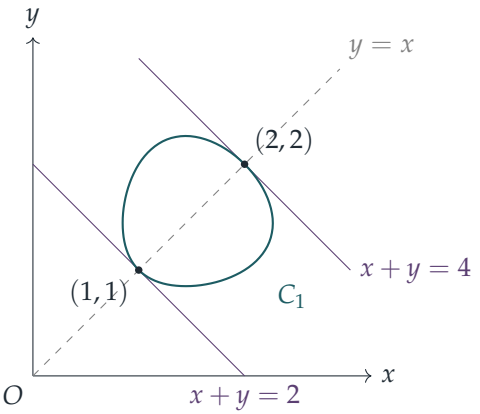
Ask yourself Is every value of s in $[2, 4]$ actually reached, and what does the curve look like at the ends?

For each $s \in [2, 4]$, put $d = \pm\sqrt{s^2 - 2^s}$ and $x = \frac{1}{2}(s + d)$, $y = \frac{1}{2}(s - d)$. Since $2^s > 0$, we have $|d| < s$, so x and y are both positive. So every such s gives a point of C_1, two points when $2 < s < 4$. The curve is therefore a closed loop:

- At $s = 2$ and $s = 4$ we have $d = 0$, giving the points $(1, 1)$ and $(2, 2)$ on $y = x$. Both points satisfy $kxy - y = -(x - kxy)$, so the gradient there is -1. The loop touches the lines $x + y = 2$ and $x + y = 4$ at these points.
- The gradient is 0 where $x = 1/\ln 2 \approx 1.44$ and infinite where $y = 1/\ln 2$. At $x = 1/\ln 2$ the loop has $y \approx 0.85$ (its lowest point) and $y \approx 2.27$ (its highest).

- Its greatest width across $y = x$ is where $s^2 - 2^s$ is largest, at $s \approx 3.21$, where $|d| \approx 1.02$.

The loop is an oval, symmetric about $y = x$, squeezed between the two lines:



Part (iii)

Now $s = x + y < 0$. Write $t = -s > 0$. As before, we need

$$s^2 \geq 2^{s+4} \iff t^2 \geq 2^{4-t} \iff t^2\, 2^t \geq 16.$$

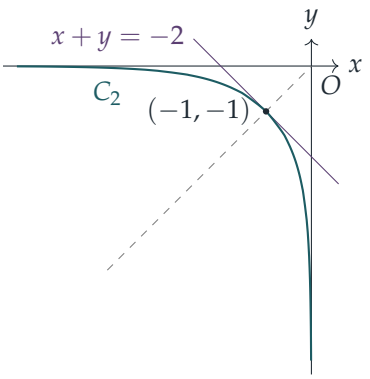
The function $t^2\, 2^t$ is increasing for $t > 0$ and equals 16 at $t = 2$. So the condition is $t \geq 2$, that is $x + y \leq -2$. (Equivalently, on a sketch of $y = x^2$ and $y = 2^{x+4}$ for $x < 0$, the parabola is above only when $x \leq -2$.) For every $s \leq -2$ the same construction as in (ii) gives two points with $x, y < 0$, except at $s = -2$, which gives the single point $(-1, -1)$. So C_2 is unbounded this time:

- It meets $y = x$ only at $(-1, -1)$, where the gradient is again -1, so it touches the line $x + y = -2$ there.
- For $x, y < 0$ we have $kx - 1 < 0$ and $1 - ky > 0$, so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y(kx - 1)}{x(1 - ky)} < 0,$$

  since the numerator is positive and the denominator negative. So y decreases steadily as x increases.
- From $xy = 4 \cdot 2^{x+y}$: as $x \to -\infty$, the product xy tends to 0, so $y \to 0^-$ (in fact $y \approx 4 \cdot 2^x / x$). By symmetry, $y \to -\infty$ as $x \to 0^-$. The negative axes are asymptotes.

So C_2 looks like one branch of a rectangular hyperbola in the third quadrant, but it hugs the axes much more tightly:



Checks

- Differentiate $4xy = 2^{x+y}$ implicitly: $4y + 4xy' = k\,2^{x+y}(1 + y') = 4kxy(1 + y')$. Rearranging gives $y' = \dfrac{kxy - y}{x - kxy}$, the original equation. The same works for C_2.
- $(2, 2)$ lies on C_1: $4 \cdot 2 \cdot 2 = 16 = 2^4$. On C_2, the point with $x + y = -4$ is about $(-3.936, -0.064)$, and $xy \approx 0.252$ while $4 \cdot 2^{-4} = 0.25$. The extreme points of the oval quoted above were confirmed by computer.

Common slips

- Forgetting the modulus signs when integrating $1/x$ and $1/y$. For C_2 they matter, and the constant comes out as $A = 4$, not $\frac{1}{4}$.
- Showing only $(x + y)^2 \geq 2^{x+y}$ and stopping. You must also explain why this confines $x + y$ to $[2, 4]$, which is where the sketch of x^2 against 2^x comes in.
- Drawing C_2 crossing the axes. On C_2 both coordinates stay negative, and the axes are asymptotes.

Beyond STEP

Rotating coordinates to $s = x + y$, $d = x - y$ turns a symmetric equation into "$d^2 = f(s)$". The curve then exists exactly where $f(s) \geq 0$: it is a closed loop when that set is a bounded interval and unbounded when it is a half-line. The same reasoning draws phase portraits from a conserved quantity: $\ln(xy) - k(x + y)$ is constant along every solution, just as energy is in the Lotka–Volterra predator–prey equations, whose orbits are closed loops for the same reason. Boyce and DiPrima, *Elementary Differential Equations*, covers both ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.